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Orientation-Dependent Universality of Spin-Transport Fluctuations in Two-Dimensional Altermagnets

Yihan Zhang,[1] Zheng Qin,[1] Donghui Xu,[1,2] Junjie Zeng,[1,*] Zhen Ning,[1,†] and Rui Wang[1,2,‡]

[1]*Institute for Structure and Function & Department of Physics & Chongqing Key Laboratory for Strongly Coupled Physics, Chongqing University, Chongqing 400044, P. R. China*

[2]*Center of Quantum Materials and Devices, Chongqing University, Chongqing 400044, P. R. China*

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Universal conductance fluctuations are conventionally organized by symmetry class, but crystalline anisotropy introduces additional structure in spin transport. Here we demonstrate orientation-dependent universal spin-transport fluctuations in disordered two-dimensional d-wave altermagnets without spin-orbit coupling. Rotating the altermagnetic orientation changes both longitudinal spin-conductance and transverse spin-Hall conductance fluctuations, while each conserved-spin sector remains in class AI. Altermagnetic orientation controls both the fluctuation amplitudes and the development of disorder-insensitive plateaus. These plateaus become flatter and less sensitive to the altermagnetic strength over the studied parameter ranges as the orientation approaches 45°, with representative fluctuation amplitudes of approximately $0.60e/(4\pi)$ for the longitudinal spin conductance and $0.40\,e/(4\pi)$ for the spin-Hall conductance. The longitudinal fluctuation amplitude correlates strongly with anisotropy of the clean-band Fermi-velocity tensor. Spin-sector covariances explain how the conserved channels combine into total fluctuations. Our results extend anisotropy-sensitive universal-fluctuation physics to conserved-spin altermagnetic transport and establish crystalline orientation as a control parameter within a fixed symmetry class.

I. INTRODUCTION

Conductance fluctuations arise from quantum interference between different scattering paths in a phase-coherent conductor. In the diffusive regime, their root-mean-square amplitude can become nearly independent of disorder strength and remain of order e^2/h. This phenomenon, known as universal conductance fluctuations (UCF), was established theoretically and observed experimentally [1–5]. The Wigner–Dyson symmetry class provides a central framework for understanding UCF, but it does not uniquely fix the fluctuation amplitude in an arbitrary device. Dimensionality, sample geometry, boundary conditions, and channel degeneracy also enter [6, 7]. Magnetic perturbations and spin-dependent scattering further reveal the role of coherent channels and their correlations [8–11]. Related studies have explored conductance fluctuations in ballistic cavities, magnetic fields, metal–insulator crossovers, and systems with superconducting contacts [12–17]. Within this broader picture, band anisotropy offers a way to modify fluctuations without necessarily changing the symmetry class. Anisotropic diffusion changes the interference modes supported by a finite sample and makes their contributions sensitive to the transport direction. Gate-dependent UCF amplitudes in Cd_3As_2 have been attributed to Fermi-surface anisotropy, while recent theoretical work has examined the effects of anisotropic bands directly [18, 19]. A related directional sensitivity occurs in spin-orbit-coupled quantum wires when the in-plane magnetic field is rotated [20]. These developments motivate extending the study of anisotropy-sensitive fluctuations from charge transport to spin transport.

Spin-conductance fluctuations contain information about both spin-resolved transmission and correlations between transport channels. When the measured spin component is conserved, the longitudinal spin conductance is proportional to the difference between spin-up and spin-down transmissions. Its variance therefore involves the variance of each channel and their covariance. The absence of spin mixing does not make the channels statistically independent, because they experience the same disorder configuration. Studies of magnetic graphene nanoribbons illustrate how spin-dependent transmission can produce substantial sample-to-sample fluctuations, even when the ensemble-averaged spin conductance vanishes [21, 22]. Transverse spin transport provides a complementary setting. The spin Hall effect has established an important route to electrical spin-current generation [23–30], with mesoscopic spin currents described through spin-resolved transport in multiterminal devices [31]. In disordered systems with Rashba or Dresselhaus spin-orbit coupling, spin-Hall conductance fluctuations can develop disorder-insensitive plateaus [32, 33]. Related studies have examined magnetic-field-dependent spin Hall transport [34] and characteristic fluctuation amplitudes in graphene and chaotic Dirac quantum dots [35, 36]. Geometric correlations can also generate a finite average spin conductance, showing that mesoscopic spin transport retains information about the underlying classical dynamics [37]. Beyond spin-orbit-driven mechanisms, anisotropic ferromagnets can support an exchange-driven transverse spin current without spin-orbit coupling [38]. These findings lead to a natural question: how does intrinsic band anisotropy control longitudinal and transverse spin-conductance fluctuations when spin remains conserved?

Here we address this question in disordered two-dimensional d-wave altermagnets without spin-orbit coupling. Altermagnets combine compensated collinear magnetic order with momentum-dependent spin splitting [39–42]. In a d-

* jjzeng@cqu.edu.cn
† zhenning@cqu.edu.cn
‡ rcwang@cqu.edu.cn

wave altermagnet, the spin splitting changes sign with momentum direction, giving the two conserved-spin sectors distinct anisotropic dispersions. This connects spin transport to the orientation of the crystalline axes relative to the device. Interest in this connection has been stimulated by studies of spin conversion and torque [43–45], spectroscopic signatures and proposed optical probes [46–49], and thermal, Hall, and superconducting responses [50–54]. Recent work has also developed descriptions of spin relaxation and spin-transfer transport in these systems [55, 56]. Using a disordered lattice model, we calculate longitudinal spin-conductance fluctuations in a two-terminal device and transverse spin-Hall conductance fluctuations in a four-terminal device. We find that crystalline orientation controls both the fluctuation amplitudes and the development of disorder-insensitive plateaus, while each conserved-spin sector remains in class AI. As the orientation approaches $\theta = \pi/4$, the plateaus become flatter and less sensitive to the altermagnetic strength over the parameter ranges studied. Representative longitudinal and transverse amplitudes reach approximately $0.60\,e/(4\pi)$ and $0.40\,e/(4\pi)$, respectively. The longitudinal fluctuation amplitude quantitatively tracks the clean-band transport anisotropy, characterized by the Fermi-velocity tensor. A spin-sector covariance analysis further identifies how correlations between the conserved channels enter the total fluctuations. Our results extend anisotropy-sensitive fluctuation physics to conserved-spin altermagnetic transport and establish crystalline orientation as a control parameter within a fixed symmetry class.

II. MODEL AND METHOD

A. Two- and four-terminal geometries and Hamiltonian

We consider a square central region of size L, with the lattice model and its spin-split electronic structure shown in Fig. 1. Two transport geometries are used, as illustrated later in Fig. 2(a) and 2(b). The two-terminal device contains two opposing altermagnetic leads and probes the longitudinal spin conductance. The four-terminal Hall bar retains these longitudinal leads and adds two normal, spin-conserving transverse probes. The altermagnetic longitudinal leads match the spin-split propagating modes of the sample, while the normal transverse leads resolve the spin-Hall response.

The disordered central region is described by a spinful tight-binding Hamiltonian

$$H = -t \sum_{\langle ij \rangle} c_i^\dagger c_j + \sum_i \varepsilon_i c_i^\dagger c_i + H_{\mathrm{AM}}, \tag{1}$$

where $c_i^\dagger = (c_{i\uparrow}^\dagger, c_{i\downarrow}^\dagger)$, t is the nearest-neighbor hopping amplitude, and ε_i is the onsite potential. In the disordered sample, ε_i is drawn independently from the uniform Anderson interval $[-W/2, W/2]$, while in the ideal leads it is set to zero.

Following the d-wave altermagnetic lattice model of Ref. [52, 55], the altermagnetic contribution is implemented

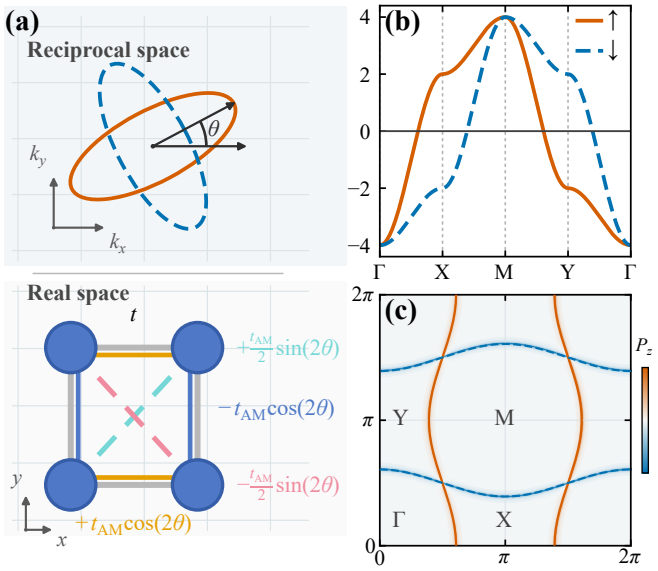

FIG. 1. Schematic illustration of the tight-binding model described by Eq. (1) and its spin-split electronic structure. (a) The upper panel illustrates the orientation of the altermagnetic anisotropy and the associated angle θ in reciprocal space. The lower panel shows the corresponding real-space hopping pattern. (b) Spin-resolved band structure along the high-symmetry path Γ-X-M-Y-Γ. The horizontal line marks the Fermi energy $E_{\mathrm{F}} = 0$. The orange solid and blue dashed curves represent the spin-up and spin-down channels, respectively. (c) Spin-resolved momentum-space spectral weight at $E_{\mathrm{F}} = 0$, shown in the equivalent Brillouin-zone representation $0 \le k_x, k_y \le 2\pi$. The color hue represents the spin polarization $P_z = (A_\uparrow - A_\downarrow)/(A_\uparrow + A_\downarrow)$, while the opacity reflects the total spectral weight $A_\uparrow + A_\downarrow$. For the numerical results in panels (b) and (c), we set $\varepsilon_0 = 0$, $t_{\mathrm{AM}} = 0.5t$, and $\theta = 0$. The nearest-neighbor hopping t is used as the energy unit, and a Lorentzian broadening of $\eta = 0.032t$ is used in panel (c).

through anisotropic spin-dependent hopping terms,

$$H_{\mathrm{AM}} = t_{\mathrm{AM}} \sum_i \Big[\cos(2\theta)(c_i^\dagger s_z c_{i+\hat{x}} - c_i^\dagger s_z c_{i+\hat{y}}) + \frac{\sin(2\theta)}{2}(c_i^\dagger s_z c_{i+\hat{x}+\hat{y}} - c_i^\dagger s_z c_{i+\hat{x}-\hat{y}}) + \mathrm{H.c.}\Big], \tag{2}$$

where t_{AM} sets the altermagnetic strength, θ is the angle between the crystalline anisotropy axis and the transport direction, and s_z is the Pauli matrix in spin space. Because only s_z appears, the transport problem remains block diagonal in the spin basis used here. This exact spin conservation is the key distinction from conventional spin-orbit-coupled spin-Hall models.

In the clean limit, the Hamiltonian takes the compact momentum-space form

$$\mathcal{H}(\boldsymbol{k}) = \varepsilon(\boldsymbol{k}) s_0 + m_{\mathrm{AM}}(\boldsymbol{k}, \theta) s_z, \tag{3}$$

where

$$\varepsilon(\boldsymbol{k}) = -2t(\cos k_x + \cos k_y), \tag{4}$$

and

$$m_{\mathrm{AM}}(\boldsymbol{k}, \theta) = 2t_{\mathrm{AM}}\big[\cos(2\theta)(\cos k_x - \cos k_y) - \sin(2\theta)\sin k_x \sin k_y\big]. \tag{5}$$

Thus the two conserved spin sectors have dispersions

$$E_\sigma(\boldsymbol{k}) = \varepsilon(\boldsymbol{k}) + \sigma m_{\mathrm{AM}}(\boldsymbol{k},\theta), \qquad \sigma = \pm 1. \tag{6}$$

An instructive limiting case occurs at $\theta = 0$, where the effective nearest-neighbor hoppings are $t_{x,\sigma} = t - \sigma t_{\mathrm{AM}}$ and $t_{y,\sigma} = t + \sigma t_{\mathrm{AM}}$. At $t_{\mathrm{AM}} = t$, the $\sigma = +1$ sector has $t_{x,+} = 0$ and is therefore completely blocked for transport along x. This exact channel-removal limit illustrates how strongly the crystalline orientation can reshape the transport ensemble without introducing spin mixing.

This form makes explicit that the altermagnetic term produces a momentum-dependent spin splitting without spin mixing. In the vicinity of the Γ point,

$$m_{\mathrm{AM}}(\boldsymbol{k},\theta) \approx -t_{\mathrm{AM}}\cos(2\theta)(k_x^2 - k_y^2) - 2t_{\mathrm{AM}}\sin(2\theta)k_x k_y, \tag{7}$$

which displays the rotated d-wave form near Γ. On the full lattice, varying θ changes the relative weights of the two lattice harmonics and can modify both the Fermi-surface shape and the strength of the transport anisotropy. Thus, away from the continuum limit, this parameterization is not generally a rigid rotation of a fixed band structure.

For scalar onsite disorder, each conserved-spin block is real in the site basis and admits complex conjugation as an effective antiunitary symmetry with square +1. Each block therefore belongs to class AI for every θ [57]. This effective symmetry within a spin block is distinct from physical spinful time reversal, which exchanges the two spin sectors.

B. Transport observables and disorder averaging

The transport calculations are carried out with the nonequilibrium Green's-function method and the Landauer-Büttiker scattering formalism [58–61]. The retarded Green's function of the open sample is

$$G^{\mathrm{r}}(E) = \Big[E\mathrm{I} - H_{\mathrm{C}} - \sum_p \Sigma_p^{\mathrm{r}}(E)\Big]^{-1}, \tag{8}$$

with H_{C} the Hamiltonian of the finite central region and Σ_p^{r} the lead self-energy of terminal p, which can be evaluated from the surface Green's function of the corresponding semi-infinite lead [62, 63]. Since spin is conserved, the scattering matrix is block diagonal in the spin-up and spin-down sectors. The two blocks are dynamically decoupled but sample the same disorder configuration and need not be statistically independent.

The spin-resolved transmission probability entering the Landauer-Büttiker formula is obtained from the Caroli, or equivalently Fisher-Lee, expression [60, 64, 65]

$$T_{ps,q}(E) = \mathrm{Tr}\Big[\Gamma_{p,s}(E)G_s^{\mathrm{r}}(E)\Gamma_{q,s}(E)G_s^{\mathrm{a}}(E)\Big], \tag{9}$$

where $G_s^{\mathrm{a}} = (G_s^{\mathrm{r}})^\dagger$ and $\Gamma_{p,s}(E) = \mathrm{i}[\Sigma_{p,s}^{\mathrm{r}}(E) - \Sigma_{p,s}^{\mathrm{a}}(E)]$ is the linewidth matrix of terminal p in spin sector s.

We take $e > 0$ and define the longitudinal bias by $V = (\mu_0 - \mu_1)/e$, where μ_q is the electrochemical potential of lead q. Spin current is positive for positive z-angular momentum flowing from the sample into the receiving lead. In both geometries, channel conductances are expressed in spin-conductance units without incorporating the sign of the spin angular momentum; that sign enters when the two channels are combined. In the two-terminal geometry, we therefore define the longitudinal channel contributions for $\alpha = \uparrow, \downarrow$ as

$$\sigma_\alpha = \frac{e}{4\pi} T_{1\alpha,0}, \tag{10}$$

and the longitudinal spin conductance is

$$\sigma_s = \sigma_\uparrow - \sigma_\downarrow = \frac{e}{4\pi}\left(T_{1\uparrow,0} - T_{1\downarrow,0}\right). \tag{11}$$

Here $\sigma_s = I_1^z/V$ denotes the total two-terminal spin conductance, with I_1^z the spin current entering lead 1. The opposite spin angular momenta are accounted for by the minus sign between the two channel contributions, exactly as in the four-terminal definition below.

In the four-terminal geometry, we use symmetric longitudinal bias, $\mu_0 = \mu_{\mathrm{eq}} + eV/2$ and $\mu_1 = \mu_{\mathrm{eq}} - eV/2$, with the transverse terminals held at $\mu_2 = \mu_3 = \mu_{\mathrm{eq}}$. The antisymmetric spin-Hall conductance is $\sigma_{\mathrm{sH}} = (I_2^z - I_3^z)/(2V)$, where I_p^z is the spin current entering transverse lead p. For each spin channel $s = \uparrow, \downarrow$, we define the transverse channel contribution in the same convention as Eq. (10),

$$\sigma_{\mathrm{H},s} = \frac{e}{16\pi}\Big[(T_{2s,0} - T_{2s,1}) - (T_{3s,0} - T_{3s,1})\Big], \tag{12}$$

where $T_{ps,q}$ is the transmission probability from longitudinal lead $q = 0, 1$ into transverse lead $p = 2, 3$ in spin sector s; because spin is conserved, the incoming-spin label is implicit. The factor $1/4$ relative to $e/4\pi$ combines the symmetric bias and the half-difference of the two transverse currents. The signs within each channel describe this spatial Hall combination; the opposite spin angular momenta enter through the channel difference in the total spin-Hall conductance,

$$\begin{aligned}\sigma_{\mathrm{sH}} &= \sigma_{\mathrm{H},\uparrow} - \sigma_{\mathrm{H},\downarrow} \\ &= \frac{e}{16\pi}\Big[(T_{2\uparrow,0} - T_{2\downarrow,0}) - (T_{2\uparrow,1} - T_{2\downarrow,1}) \\ &\qquad - (T_{3\uparrow,0} - T_{3\downarrow,0}) + (T_{3\uparrow,1} - T_{3\downarrow,1})\Big].\end{aligned} \tag{13}$$

In what follows, all channels contributions and total spin conductance are reported in units of $e/4\pi$.

Disorder averaging is performed at fixed $(E, W, t_{\mathrm{AM}}, \theta, L)$. Details of the ensemble construction and numerical consistency checks are provided in Appendix A. For an observable X, we use

$$\langle X\rangle = \frac{1}{N_{\mathrm{s}}}\sum_{n=1}^{N_{\mathrm{s}}} X_n, \tag{14}$$

and define the fluctuation amplitude for either total spin response $Y = \sigma_{\mathrm{s}}, \sigma_{\mathrm{sH}}$ as

$$\mathrm{rms}(Y) = \sqrt{\langle Y^2\rangle - \langle Y\rangle^2}. \tag{15}$$

To compare the two geometries at the spin-channel level, let $Z_\alpha = \sigma_\alpha$ for the longitudinal device and $Z_\alpha = \sigma_{\mathrm{H},\alpha}$ for the Hall device, with $\alpha = \uparrow, \downarrow$. In both cases, the measured spin conductance is $Y = Z_\uparrow - Z_\downarrow$. We evaluate

$$\mathrm{cov}(Z_\uparrow, Z_\downarrow) = \langle Z_\uparrow Z_\downarrow \rangle - \langle Z_\uparrow \rangle \langle Z_\downarrow \rangle, \tag{16}$$

and the corresponding correlation coefficient

$$\rho_{\uparrow\downarrow} = \frac{\mathrm{cov}(Z_\uparrow, Z_\downarrow)}{\sqrt{\mathrm{var}(Z_\uparrow)\,\mathrm{var}(Z_\downarrow)}}. \tag{17}$$

Because both measured spin responses are differences between channel contributions, their variance obeys

$$\begin{aligned} \mathrm{var}(Y) &= \mathrm{var}(Z_\uparrow) + \mathrm{var}(Z_\downarrow) - 2\,\mathrm{cov}(Z_\uparrow, Z_\downarrow) \\ &= \mathrm{var}(Z_\uparrow) + \mathrm{var}(Z_\downarrow) - 2\,\rho_{\uparrow\downarrow}\sqrt{\mathrm{var}(Z_\uparrow)\,\mathrm{var}(Z_\downarrow)}, \end{aligned} \tag{18}$$

with all variances and covariances expressed in units of $(e/4\pi)^2$.

Energies, disorder strengths, and altermagnetic couplings are measured in units of the nearest-neighbor hopping t. The numerical implementation is carried out with the Kwant package [61].

III. RESULTS

A. Energy dependence of the average response

We first examine the disorder-averaged transport response in the two geometries introduced above. Figures 2(c) and 2(d) show the energy dependence of the averaged longitudinal spin conductance at $\theta = 0$ and the spin-Hall conductance at $\theta = \pi/4$, respectively. In both geometries, increasing disorder rapidly suppresses the disorder-averaged response. This decay of the average signals provides the background against which the much more robust mesoscopic fluctuations are identified.

B. Angular anisotropy of the average response

The angular scans in Fig. 2(e) and 2(f) display the direct transport imprint of the rotated d-wave altermagnetic texture. The two-terminal spin conductance is extremal close to $\theta = \pi/2$ and changes sign near $\theta = \pi/4$ and $3\pi/4$. The four-terminal spin-Hall signal is enhanced near $\theta \approx \pi/4$, reverses sign around $\theta = \pi/2$, and exhibits the opposite extremum near $3\pi/4$. Increasing disorder reduces the magnitude of both signals without removing their symmetry-imposed angular structures. The persistence of this orientation dependence after disorder averaging shows that the mean transport response retains the anisotropy of the altermagnetic model. We next examine the corresponding conductance-fluctuation statistics.

C. Anisotropy-selected universal fluctuation values

We now turn to the mesoscopic fluctuations. Figure 3 shows the root-mean-square fluctuations of the longitudinal and transverse spin observables. The upper row gives the two-terminal spin-conductance fluctuation $\mathrm{rms}(\sigma_s)$, whereas the lower row gives the four-terminal spin-Hall fluctuation $\mathrm{rms}(\sigma_{\mathrm{sH}})$. The focus is the orientation dependence of the fluctuation amplitudes and the development of approximately disorder-insensitive plateaus within class AI.

The energy scans in Fig. 3(a) and 3(d) are performed at $\theta = \pi/4$. The longitudinal spin-conductance fluctuation forms a broad central-band plateau whose scale is close to $0.6\,e/4\pi$ for weak and intermediate disorder, and decreases near the band edges. The spin-Hall fluctuation stays near $0.4\,e/4\pi$ over much of the central energy window. Although the detailed structures differ, both observables display broad energy intervals in which their amplitudes depend only weakly on energy. We consequently choose $E = t$ for the orientation and disorder scans.

The angular scans in Fig. 3(b) and 3(e) directly test whether the fluctuation amplitudes remain sensitive to crystalline ori-

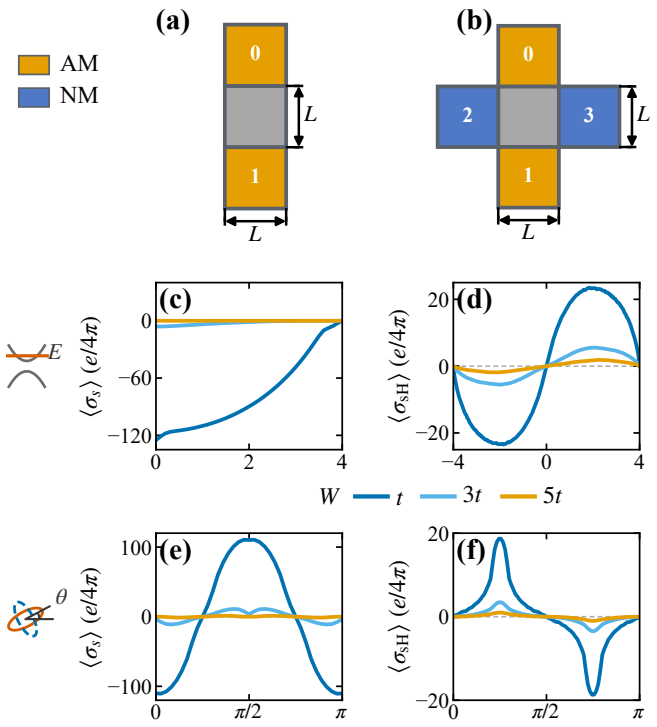

FIG. 2. Two- and four-terminal transport geometries and their corresponding disorder-averaged responses. (a,b) Two- and four-terminal geometries, respectively. The orange and blue regions denote altermagnetic (AM) and normal-metal (NM) leads, while the gray square represents the disordered scattering region of size $L \times L$. The numbers label the lead indices. (c,e) Disorder-averaged spin conductance $\langle \sigma_s \rangle$ for the two-terminal geometry shown in panel (a), as functions of the Fermi energy E and the altermagnetic orientation angle θ, respectively. (d,f) Corresponding spin Hall conductance $\langle \sigma_{\mathrm{sH}} \rangle$ for the four-terminal geometry shown in panel (b). For the energy scans, $\theta = 0$ in panel (c) and $\theta = \pi/4$ in panel (d); for the angular scans in panels (e,f), $E = t$. The curves represent disorder strengths $W = t$, $3t$, and $5t$. All calculations use $L = 400$ and $t_{\mathrm{AM}} = 0.9t$, with the nearest-neighbor hopping t taken as the energy unit. Each data point is averaged over 3000 disorder realizations.

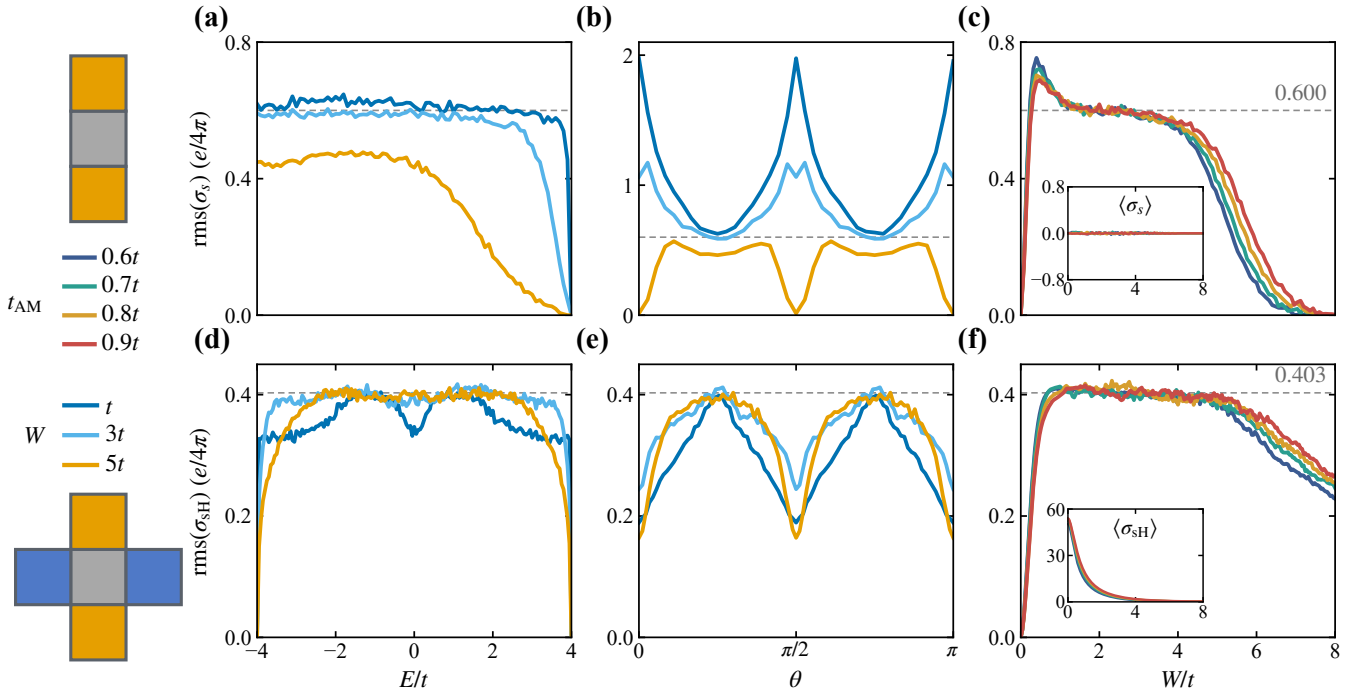

FIG. 3. Mesoscopic conductance fluctuations in the two- and four-terminal geometries at $L = 400$. The device schematics on the left identify the geometry associated with each row. (a–c) Spin-conductance fluctuation rms(σ_s) in the two-terminal geometry. (d–f) Spin-Hall conductance fluctuation rms(σ_{sH}) in the four-terminal geometry. Panels (a,d), (b,e), and (c,f) show the dependence on the Fermi energy E, the altermagnetic orientation angle θ, and the disorder strength W, respectively. For the energy and angular scans in panels (a,b,d,e), $t_{AM} = 0.9t$ and $W = t$, $3t$, and $5t$. The energy scans are calculated at $\theta = \pi/4$, while the angular scans are calculated at $E = t$. For the disorder scans in panels (c,f), $E = t$ and $\theta = \pi/4$, with $t_{AM} = 0.6t$, $0.7t$, $0.8t$, and $0.9t$. The gray dashed lines mark the reference fluctuation amplitudes $0.600\, e/4\pi$ and $0.403\, e/4\pi$ for the two- and four-terminal geometries, respectively. The insets in panels (c,f) show the corresponding disorder-averaged conductances $\langle\sigma_s\rangle$ and $\langle\sigma_{sH}\rangle$. The nearest-neighbor hopping t is used as the energy unit, and each data point is averaged over 3000 disorder realizations.

entation within the same class-AI symmetry. The two-terminal fluctuation follows a pronounced periodic angular pattern, with minima at specific crystalline orientations, while the spin-Hall fluctuation develops broad maxima near $\theta = \pi/4$ and $3\pi/4$. Throughout these rotations each fixed-spin block remains real and stays in class AI; there is no orthogonal-to-unitary or orthogonal-to-symplectic crossover. The angular dependence therefore occurs within a fixed symmetry class and reflects the anisotropic transport structure of the model. This illustrates how crystalline orientation can change the fluctuation scale without a symmetry-class crossover.

Finally, Fig. 3(c) and 3(f) show the evolution of the fluctuations as a function of disorder strength W at $E = t$ and $\theta = \pi/4$. Both fluctuations rise rapidly from the clean limit, become nearly independent of W and t_{AM} over broad intyermediate-disorder intervals, and decrease at stronger disorder. The dashed lines indicate the representative levels rms(σ_s) $\approx 0.600\, e/4\pi$ for the longitudinal spin conductance and rms(σ_{sH}) $\approx 0.403\, e/4\pi$ for the spin-Hall conductance. Averaging over the plateau windows for $t_{AM} = 0.6t$–$0.9t$ gives $0.595 \pm 0.009\, e/4\pi$ and $0.399 \pm 0.006\, e/4\pi$, respectively, where the quoted spreads quantify the residual variation across W and t_{AM} within the plateau region. To examine whether the angular differences persist across a disorder interval, we compare the longitudinal disorder scans at $\theta = \pi/12, \pi/6$, and $\pi/4$(see Appendix C and Fig. 6). At $t_{AM} = 0.9t$,averaging the fluctuation amplitudes over $1.6 \le W/t \le 3.2$ gives approximately $0.87e/4\pi, 0.66e/4\pi$ and $0.60e/4\pi$, respectively. The separated amplitudes persist over this common window, with stronger residual disorder dependence at $\pi/12$. Thus the orientation dependence is not confined to a single disorder point.

The analogous four-terminal disorder scans, performed at $\theta = \pi/12$, $\pi/6$, and $\pi/4$, reveal a closely related orientation dependence. For $t_{AM} = 0.6t$–$0.9t$, the intermediate-disorder spin-Hall fluctuation evolves from an average scale of approximately $0.353\, e/4\pi$ at $\theta = \pi/12$ to $0.381\, e/4\pi$ at $\theta = \pi/6$ and $0.399\, e/4\pi$ at $\theta = \pi/4$. More importantly, as the crystalline axis approaches $\theta = \pi/4$, the fluctuation curves become progressively flatter as functions of W and collapse more tightly across different t_{AM} values (see Appendix C and Fig. 7). These three representative orientations show the same qualitative evolution as the two-terminal results: the fluctuation plateau becomes increasingly well developed toward $\theta = \pi/4$.

A microscopic measure of this evolution is supplied by the eigenvalue ratio η_v of the spin-resolved Fermi-surface velocity tensor. At $E = t$ and $t_{AM} = 0.9t$, η_v decreases from 25.6 at $\theta = \pi/12$ to 4.92 at $\pi/6$ and 1.70 at $\pi/4$. Across the irreducible angular interval, the longitudinal fluctuation at $W = 3t$ has a correlation coefficient of 0.956 with $\ln \eta_v$ (see Appendix D and Fig. 8). The fixed-W angular scan therefore shows a strong correlation with clean-band velocity anisotropy. The longitudinal plateau amplitudes follow the same decreasing

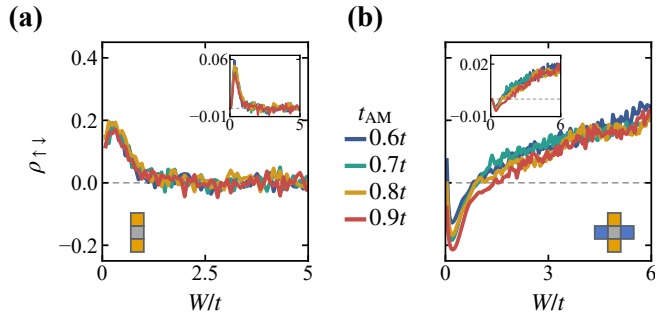

FIG. 4. Disorder dependence of the correlation between the two conserved spin channels. (a) Correlation coefficient $\rho_{\uparrow\downarrow}$ between the spin-up and spin-down longitudinal conductances in the two-terminal geometry. (b) Corresponding correlation coefficient between the spin-resolved Hall conductances in the four-terminal geometry. The correlation coefficient is defined in Eq. (17). The insets show the corresponding covariance, $\mathrm{cov}(\sigma_\uparrow, \sigma_\downarrow)$ in panel (a) and $\mathrm{cov}(\sigma_{\mathrm{H},\uparrow}, \sigma_{\mathrm{H},\downarrow})$ in panel (b), as defined in Eq. (16). The covariance is expressed in units of $(e/4\pi)^2$ in both panels. The curves correspond to $t_{\mathrm{AM}} = 0.6t$, $0.7t$, $0.8t$, and $0.9t$. The gray dashed lines indicate zero correlation or covariance, and the device symbols identify the respective transport geometries. All results are calculated at $L = 400$, $E = t$, and $\theta = \pi/4$. Each data point is averaged over 3000 disorder realizations.

trend across the three representative orientations.D The singular endpoint described above, $\theta = 0$ and $t_{\mathrm{AM}} = t$, provides a microscopic limiting case by eliminating one longitudinal spin channel.

The additional four-terminal calculations at $\theta = \pi/4$ show modest variation over $L = 400 - 600$ at the tested parameter(see Appendix Sec. B and Fig. 5). This supports the persistence of the transverse fluctuation plateau over this size range. The representative amplitude near $\theta = \pi/4$, approximately $0.40e/4\pi$, substantially exceeds the conventional USCF amplitude of $0.18e/4\pi$ reported for strong spin-orbit-coupled systems[32].

D. Disorder-driven correlations between conserved spin sectors

Figure 4 resolves the correlations that accompany the disorder crossover. In the two-terminal geometry [Fig. 4(a)], the correlation coefficient between the spin-up and spin-down contributions to the longitudinal spin conductance is positive at weak disorder, falls rapidly toward zero, and subsequently fluctuates around zero. The inset confirms that the corresponding covariance becomes small once the initial weak-disorder regime has been left. According to Eq. (18), the longitudinal spin-conductance variance then approaches the sum of the two channel variances. Thus, the conductance fluctuations of the two conserved spin sectors are approximately uncorrelated over most of the tested two-terminal fluctuation plateau.

In the four-terminal Hall geometry [Fig. 4(b)], the inter-channel correlation is negative at weak disorder and then increases continuously with W, crossing zero in the intermediate-disorder range. The covariance in the inset follows the same trend. Since $\sigma_{\mathrm{sH}} = \sigma_{\mathrm{H},\uparrow} - \sigma_{\mathrm{H},\downarrow}$, a negative covariance enhances the total spin-Hall variance, whereas a positive covariance reduces it, as is explicit in Eq. (18). This covariance is not specified by the Wigner–Dyson index of an individual spin block. The decomposition shows how inter-channel correlations alter the total variance at the displayed orientation without a change of symmetry class.

The two spin sectors share the same scalar disorder but have different spin-dependent hopping amplitudes. Their transmissions can therefore be correlated even though spin is conserved. Figure 4 shows that the sign and magnitude of this covariance depend on the transport geometry and disorder strength. Through Eq. (18), these correlations can either enhance or reduce the total spin-conductance variance.

Taken together, Figs. 3 and 4 separate two ingredients that are often closely linked in conventional isotropic treatments. Symmetry determines the interference class of each conserved spin sector, whereas altermagnetic anisotropy and the resulting inter-sector covariance shape the numerical amplitude assembled from those sectors. In the studied strong-altermagnetic regime, the plateaus are approximately independent of disorder strength, while their amplitudes depend on the anisotropic transport configuration.

IV. CONCLUSION

We have shown that crystalline anisotropy provides an additional organizing principle for universal mesoscopic spin transport in altermagnets. Extending recent evidence that band and Fermi-surface anisotropy can reshape charge UCF [18, 19], we find orientation-dependent universal fluctuations for two distinct spin observables. For the two-terminal geometry, this brings the longitudinal SCF problem previously studied in magnetic graphene nanoribbons [21, 22] to two-dimensional transport with rotatable altermagnetic anisotropy. For the four-terminal geometry, this extends the conventional USCF framework to a spin-conserving altermagnetic setting. In the present model, each conserved-spin block remains real and belongs to class AI for all orientations of the d-wave altermagnetic axis. Rotating that axis nevertheless changes both longitudinal spin-conductance and transverse spin-Hall conductance fluctuation amplitudes. Within this family of anisotropic transport ensembles, the numerical fluctuation scale is therefore not determined by the symmetry class alone.

Disorder scans at representative crystalline orientation show a systematic evolution of the fluctuation plateaus in the strong-altermagnetic regime: both longitudinal and spin-Hall fluctuations approach their clearest disorder- and coupling-insensitive plateaus near $\theta = \pi/4$, whereas stronger anisotropy away from this orientation leaves a more pronounced parameter dependence. At $\theta = \pi/4$, the representative values are $\mathrm{rms}(\sigma_s) \approx 0.60\, e/4\pi$ and $\mathrm{rms}(\sigma_{\mathrm{sH}}) \approx 0.40\, e/4\pi$. The longitudinal fluctuation amplitude is strongly correlated with the clean-band velocity anisotropy. The reduction of this anisotropy toward $\pi/4$ also accompanies the development of

latter plateaus. At the opposite extreme, $\theta = 0$ and $t_{\rm AM} = t$ eliminate one longitudinal spin channel completely. We therefore identify orientation as controlling both the fluctuation scale and how fully the universal plateau is developed within the same class-AI interference ensemble. The spin-sector covariance provides a complementary statistical description of how the conserved channels combine into the measured total fluctuations.

Experimentally, the relevant mesoscopic regime requires $\ell < L < \ell_\phi$, with a spin-relaxation length exceeding L. A relevant candidate is the metallic room-temperature d-wave altermagnet KV_2Se_2O [47], although quantitative material-specific modeling should include spin-orbit coupling, dephasing, and microscopic defects. A direct experimental test would be to compare devices oriented along different crystalline axes and search for disorder-insensitive fluctuation plateaus whose amplitudes vary systematically with orientation.

DATA AVAILABILITY

The data supporting the findings of this study are available from the corresponding authors upon request.

ACKNOWLEDGMENTS

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Appendix A: Ensemble construction and consistency checks

For every disordered transport point in the main text, the reported root-mean-square fluctuation is the population standard deviation over $N_{\rm s} = 3000$ realizations. When several independently generated batches are combined, the sample-resolved conductances are concatenated first and all averages, variances, covariances, and correlation coefficients are then recomputed from the resulting ensemble. Thus, statistics from different batches are never averaged as if they were individual samples. The clean limit $W = 0$ contains a single deterministic configuration and is consequently assigned zero disorder-induced fluctuation.

Spin-resolved quantities are evaluated for paired disorder configurations. As an internal check on both geometries, we verify the sample-by-sample identities $\sigma_s = \sigma_\uparrow - \sigma_\downarrow$ and $\sigma_{\rm sH} = \sigma_{\rm H,\uparrow} - \sigma_{\rm H,\downarrow}$ before evaluating the ensemble moments. Here $\sigma_\alpha = (e/4\pi)T_{1\alpha,0}$ for $\alpha = \uparrow, \downarrow$, and all channel contributions and total spin conductances are expressed in units of $e/4\pi$, as in the main text. For the merged four-terminal datasets, the largest absolute residual in the latter identity is below 1.1×10^{-9} in these conductance units.

Both geometries use channel quantities without the sign of the spin angular momentum; this sign is included through the up-minus-down combination. In particular, $\sigma_{\rm H,\downarrow}$ includes the spatial signs of the transverse Hall combination but no additional minus sign for the down-spin angular momentum. The channel covariances and correlation coefficients use this same convention.

To quantify the representative plateau levels, we treat every plotted parameter point inside a stated plateau window on equal footing. If $R_\alpha = {\rm rms}(Y)_\alpha$ denotes the ensemble fluctuation at the parameter point $\alpha = (W, t_{\rm AM})$, we define

$$\overline{R} = \frac{1}{N_{\rm p}} \sum_{\alpha=1}^{N_{\rm p}} R_\alpha, \qquad \Delta_{\rm pl} = \left[\frac{1}{N_{\rm p} - 1} \sum_{\alpha=1}^{N_{\rm p}} (R_\alpha - \overline{R})^2 \right]^{1/2}. \tag{A1}$$

Here $\Delta_{\rm pl}$ is the spread across the displayed W and $t_{\rm AM}$ values. It deliberately measures the residual parameter dependence of the plateau rather than the Monte Carlo sampling error of a single 3000-realization rms estimate. Here $\Delta_{\rm pl}$ is the spread across the displayed W and $t_{\rm AM}$ values. It summarizes the observed variation of the plateau estimates, including their finite-ensemble variability. It is not the Monte Carlo sampling error of a single 3000-realization rms estimate. At $\theta = \pi/4$, averaging the four $t_{\rm AM} = 0.6t$–$0.9t$ curves over $1.6 \le W/t \le 3.2$ for the longitudinal response and $2.4 \le W/t \le 4.8$ for the spin-Hall response gives

$$\overline{R}_s = 0.595 \pm 0.009 \, \frac{e}{4\pi}, \qquad \overline{R}_{\rm sH} = 0.399 \pm 0.006 \, \frac{e}{4\pi}, \tag{A2}$$

where the uncertainties are $\Delta_{\rm pl}$. These values quantify the precision of the plateaus and are consistent with the 0.600 and $0.403\, e/4\pi$ reference guides in Fig. 3 of the main text. where the ± values denote Δ_{pl}, the spread across the selected parameter points. They characterize the observed plateau variation, not the uncentainty of a fitted universal constant or the Monte Carlo error of an individual rms estimate. The 0.600 and 0.403 $e/4\pi$ lines in Fig.3 are nearby visual guides.

Appendix B: Robustness against system size and altermagnetic strength

The disorder scans in the main text show that the four-terminal spin-Hall fluctuation approaches a broad plateau near $0.40\, e/4\pi$ at fixed orientation. Here we test whether that scale is also insensitive to the system size and to moderate changes of the altermagnetic hopping. Figure 5 summarizes these additional scans, all performed at $E = t$ and $\theta = \pi/4$ with 3000 disorder realizations per point.

Figure 5(a) shows the size dependence at $t_{\rm AM} = 0.9t$. For $W = 3t$ and $5t$, ${\rm rms}(\sigma_{\rm sH})$ remains between approximately 0.396 and $0.421\, e/4\pi$ as L is varied from 400 to 600. The $W = 7t$ curve is systematically lower, consistent with the

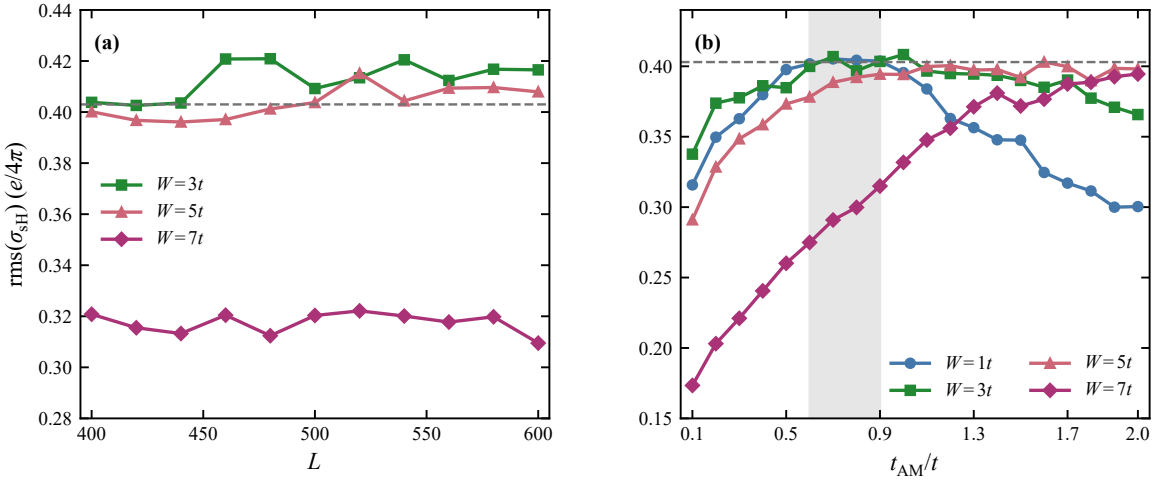

FIG. 5. Robustness of the four-terminal spin-Hall conductance fluctuation at $E = t$ and $\theta = \pi/4$. (a) System-size dependence for $t_{\mathrm{AM}} = 0.9t$ and $W = 3t$, $5t$, and $7t$. (b) Altermagnetic-coupling dependence for $L = 400$ and $W = t$, $3t$, $5t$, and $7t$. The shaded vertical band marks the interval $0.6t \le t_{\mathrm{AM}} \le 0.9t$ used for the disorder-scan comparison in the main text. The horizontal dashed line denotes $0.403\,e/4\pi$. Each point is calculated from 3000 disorder realizations.

strong-disorder suppression seen in the main-text disorder scan. The transverse fluctuation therefore remains close to $0.40e/4\pi$ over the tested range $L = 400-600$. This establishes robustness over that range rather than a thermodynamic-limit extrapolation.

Figure 5(b) displays the corresponding coupling scan at $L = 400$. In the interval $0.6t \le t_{\mathrm{AM}} \le 0.9t$ used for the main-text comparison, the $W = t$ and $3t$ curves remain within 0.397–$0.407\,e/4\pi$, while the $W = 5t$ curve approaches the same scale as t_{AM} increases. Outside this diffusive window, especially for strong disorder or weak altermagnetic splitting, the fluctuation need not remain at the plateau value. This distinction is consistent with the interpretation in the main text: universality applies within a fixed anisotropic diffusive configuration and is not asserted throughout the complete parameter space.

Appendix C: Orientation-resolved disorder crossovers

The angular scan in Fig. 3(b) of the main text shows that the longitudinal spin-conductance fluctuation varies strongly with the crystalline orientation at fixed disorder. To compare the disorder evolution at fixed orientations, we examine full disorder scans at $\theta = \pi/12, \pi/6$ and $\pi/4$. All calculations use $E = t$, $L = 400$, and $t_{\mathrm{AM}} = 0.9t$, matching the parameters of the main-text angular scan, and every point is averaged over 3000 disorder realizations.

As shown in Fig. 6, the three orientations exhibit plateau-like regions with different amplitudes and residual slopes before the strong-disorder decrease. Averaging over the shaded interval gives representative amplitudes of $0.868\,e/4\pi$, $0.662\,e/4\pi$, and $0.598\,e/4\pi$ for $\theta = \pi/12$, $\pi/6$, and $\pi/4$, respectively, with corresponding plateau-window spreads $\Delta_{\mathrm{pl}} = 0.034$, 0.008, and $0.008\,e/4\pi$. The points near $W = t, 3t$ and $5t$ are consistent with the angular dependence in the main text. The complete scans show that the separated amplitudes persist

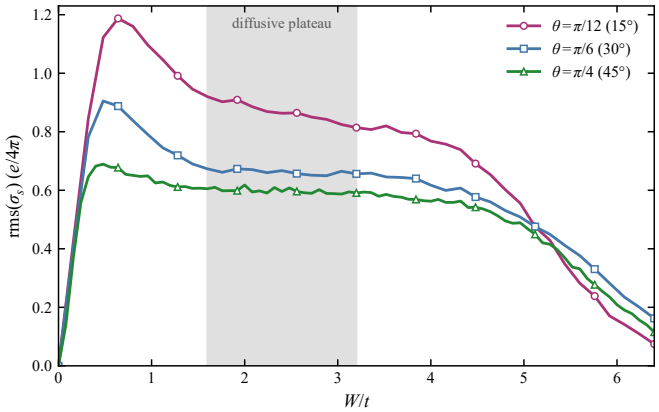

FIG. 6. Disorder dependence of the two-terminal longitudinal spin-conductance fluctuation at three representative crystalline orientations. The gray band marks the common comparison interval $1.6 \le W \le 3.2$. Each curve rises from the clean limit, develops a plateau-like region with an orientation-dependent residual slope, and decreases at stronger disorder. The calculations use $E = t$, $L = 400$, and $t_{\mathrm{AM}} = 0.9t$, with 3000 disorder realizations at every point. The $\theta = \pi/4$ curve is the corresponding dataset shown in Fig. 3(c) of the main text.

across the common comparison window, with different residual slopes. Thus, orientation affects both the average fluctuation scale and its parameter dependence in the sampled range.

We perform the corresponding four-terminal spin-Hall disorder scans at the same three orientations, now for $t_{\mathrm{AM}} = 0.6t$–$0.9t$. Figure 7 shows a continuous evolution toward the particularly clear plateau at $\theta = \pi/4$. Over the common intermediate-disorder window $2.4 \le W/t \le 4.8$, averages over both W and the four altermagnetic strengths are $0.353\,e/4\pi$, $0.381\,e/4\pi$, and $0.399\,e/4\pi$ at $\theta = \pi/12$, $\pi/6$, and $\pi/4$, respectively, with $\Delta_{\mathrm{pl}} = 0.010$, 0.009, and $0.006\,e/4\pi$. More importantly than the change in mean amplitude, the decreasing spread shows quantitatively that the curves become progressively flatter in W and collapse more tightly across t_{AM} as the orientation approaches $\pi/4$. Thus the angular dependence

of Fig. 3(e) in the main text is accompanied by a systematic strengthening of the universal plateau rather than by three equally developed plateaus. The smaller parameter spread near $\pi/4$ is consistent with flatter curves and a tighter collapse across t_{AM} over the displayed window. Here Δ_{pl} describes the aggregate variation across the sampled points, including the sampling variability of individual estimates.

Appendix D: Fermi-velocity anisotropy

To connect the orientation-dependent fluctuation scales to a microscopic transport quantity, we evaluate the spin-resolved Fermi-surface velocity tensor

$$V_{ij}^{(s)} = \frac{\displaystyle\int_{\mathrm{BZ}} \mathrm{d}^2k\, v_{i,s}(\boldsymbol{k}) v_{j,s}(\boldsymbol{k}) \delta_\gamma[E_s(\boldsymbol{k}) - E_F]}{\displaystyle\int_{\mathrm{BZ}} \mathrm{d}^2k\, \delta_\gamma[E_s(\boldsymbol{k}) - E_F]}, \qquad \delta_\gamma(x) = \frac{\gamma}{\pi(x^2+\gamma^2)}, \tag{D1}$$

where $v_{i,s} = \hbar^{-1}\partial E_s/\partial k_i$, with $a = \hbar = 1$ in the numerical evaluation, $E_{\mathrm{F}} = t, t_{\mathrm{AM}} = 0.9t$, and $\gamma = 0.032t$. We characterize the anisotropy by the coordinate-independent eigenvalue ratio $\eta_v = \lambda_{\max}(V^{(s)})/\lambda_{\min}(V^{(s)})$. The clean dispersions of the two spin blocks are related by a $\pi/2$ rotation. Their velocity tensors therefore have the same eigenvalues, so either block gives the same η_v. For nondegenrate eigenvalues, the principal axes are related by the same rotation.

Within a semiclassical relaxation-time approximation with a momentum-independent transport time, the diffusion tensor is $D_{ij}^{(s)} \simeq \tau_{\mathrm{el}}^{(s)} V_{ij}^{(s)}$. We use this relation as motivation for a clean-band anisotropy diagnostic, not as a quantitative determination of the diffusion tensor throughout the disorder range. The conductance-fluctuation correlator is a sum over eigenmodes of the diffusion operator [4]; in the present two-dimensional device these modes obey schematically

$$\left[-\partial_i D_{ij}^{(s)} \partial_j + \tau_\phi^{-1}\right] Q_\nu^{(s)} = \Lambda_\nu^{(s)} Q_\nu^{(s)}. \tag{D2}$$

The device boundaries are fixed while $D_{ij}^{(s)}$ rotates and changes its eigenvalue ratio. Consequently, both the diffusion spectrum $\{\Lambda_\nu^{(s)}\}$ and the associated fluctuation mode sum can depend on θ without a change of Wigner–Dyson class. A full analytic evaluation of the four-terminal boundary-value problem is not required for the comparison below; η_v provides a direct microscopic diagnostic of the anisotropy entering Eq. (D2).

The endpoint $\theta = 0$ provides an exact illustration of the hopping anisotropy. The spin-resolved dispersion can then be written

$$E_\sigma(\boldsymbol{k}, 0) = -2(t - \sigma t_{\mathrm{AM}})\cos k_x - 2(t + \sigma t_{\mathrm{AM}})\cos k_y. \tag{D3}$$

At $t_{\mathrm{AM}} = t$, the $\sigma = +1$ channel has no k_x dispersion, so its group velocity along the transport direction vanishes and that channel is completely blocked. The very large but finite η_v near $\theta = 0$ at $t_{\mathrm{AM}} = 0.9t$ in Fig. 8(a) is the continuous precursor of this exact channel-removal limit.

Figure 8(a) shows that η_v decreases rapidly as θ approaches $\pi/4$. In particular, $\eta_v = 25.6$, 4.92, and 1.70 at $\theta = \pi/12$, $\pi/6$, and $\pi/4$. These values evolve in parallel with the longitudinal plateau amplitudes 0.868, 0.662, and $0.598\, e/4\pi$. Over the irreducible interval $0 \le \theta \le \pi/4$, sampled every 5°, the $W = 3t$ longitudinal fluctuation and $\ln \eta_v$ have a Pearson correlation coefficient of 0.956 [Fig. 8(b)]. This comparison does not constitute an analytic formula for the universal prefactor; it establishes a strong quantitative correlation between the amplitude change and the clean-band transport anisotropy. The same reduction of η_v toward $\pi/4$ is consistent with, but does not by itself derive, the increasingly well-developed spin-Hall plateau in Fig. 7.

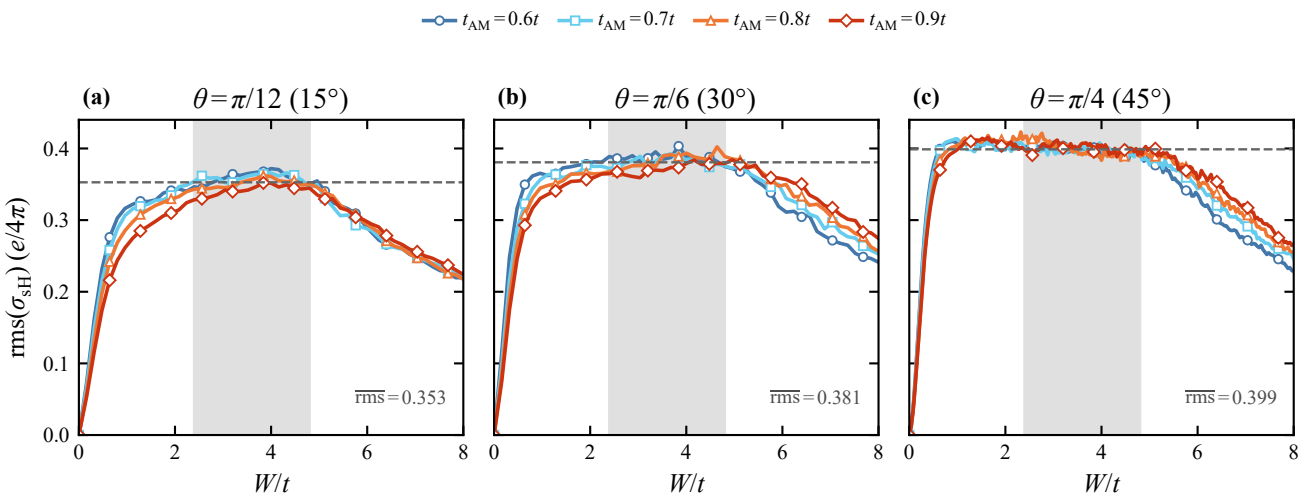

FIG. 7. Disorder dependence of the four-terminal spin-Hall conductance fluctuation at (a) $\theta = \pi/12$, (b) $\theta = \pi/6$, and (c) $\theta = \pi/4$. Each panel contains $t_{AM} = 0.6t$, $0.7t$, $0.8t$, and $0.9t$. The gray band marks the common comparison window $2.4 \le W/t \le 4.8$, and the dashed line is the average over all plotted points within that window: $0.353\,e/4\pi$, $0.381\,e/4\pi$, and $0.399\,e/4\pi$, respectively. The curves become flatter and less dependent on t_{AM} as θ approaches $\pi/4$. All calculations use $E = t$, $L = 400$, and 3000 disorder realizations per point.

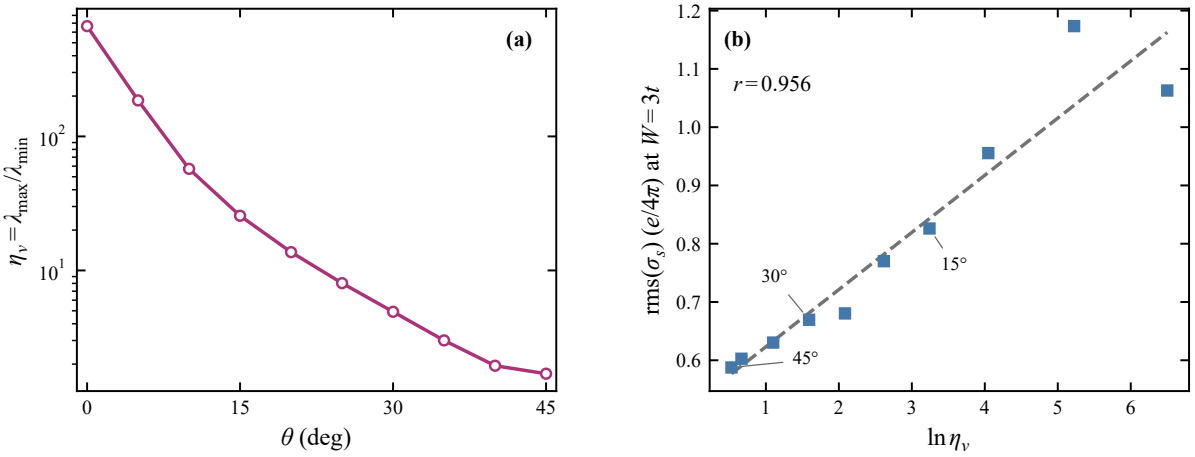

FIG. 8. Microscopic measure of transport anisotropy at $E_F = t$ and $t_{AM} = 0.9t$. (a) Eigenvalue ratio η_v of the spin-resolved Fermi-surface velocity tensor defined in Eq. (D1). (b) Longitudinal spin-conductance fluctuation at $W = 3t$ versus $\ln\eta_v$ over $0 \le \theta \le \pi/4$ in 5° increments. The dashed line is a linear guide to the eye; the Pearson correlation coefficient is $r = 0.956$.

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